

Pheno-data: knowledge from tomatoes' becoming with different ecological worlds

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(a) Vertical Farm Tomato. Image credit: Gold-locki via Wikimedia Commons



(b) Heirloom Tomato. Image purchased from Shutterstock



(c) Wild Tomato. Image credit: Kevin Gepford via Wikimedia Commons

Figure 1: Introducing our three tomatoes

ABSTRACT

Western industrialised societies are increasingly accustomed to imagining and constructing the world through a vision of data that prioritises economic, social, and technological enhancement for some humans - while sidelining the livingness of other-than-humans and the environment, which are vital for sustaining life on Earth. In this Alt.CHI paper, we introduce different tomato entities to help us explore new ways of looking at data. From an other-than-human perspective, we consider how a greenhouse tomato, an heirloom tomato, and a wild tomato perceive and embody data differently in the world, deriving lessons from these distinct viewpoints within the field of HCI.

CCS CONCEPTS

• Human-centered computing → HCI theory, concepts and models.

KEYWORDS

Pheno-data, Organisms, Livingness, More-than-Human, Ecologies

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1 INTRODUCING OUR THREE TOMATOES

In this paper, we introduce three varieties and species of tomatoes: greenhouse tomatoes, heirloom tomatoes, and wild tomatoes. By recontextualising these tomatoes in their environments, we imagine how they might have physical attributes, behaviours, and evolving characteristics as they interact with humans, other-than-humans¹, and their surroundings differently. Let's delve into their life stories.

1.1 Indoor farm tomatoes at a university of agricultural research, the UK (Figure 2)

This is a flawless, green, and firm tomato, hanging from the neat and long vine of a tomato plant. The vine is intertwined with other tomato plants, resembling electric cables, while the shoot tip of the tomato plant grows vertically along a human-made wire towards the glass ceiling. The seed of this tomato plant has been modified to resist disease, ripen evenly, and thrive efficiently in reduced light conditions, compared to its open-field counterparts. It is nurtured within a small soilless container filled with rock-based inorganic fibre substrates. Additionally, it is connected to an irrigation system that meticulously waters each plant with a tailor-made blend of nutrients, providing precise nourishment. For

¹We deliberately use the term "other-than-humans" over the more common "more-than-humans" or even "non-humans". This choice is made, in the first case, to avoid a hierarchy that obscures the fact that many communities continue to be treated as "sub-" or "less-" than human, and, in the second case to avoid the human-centric of either being or not being human.

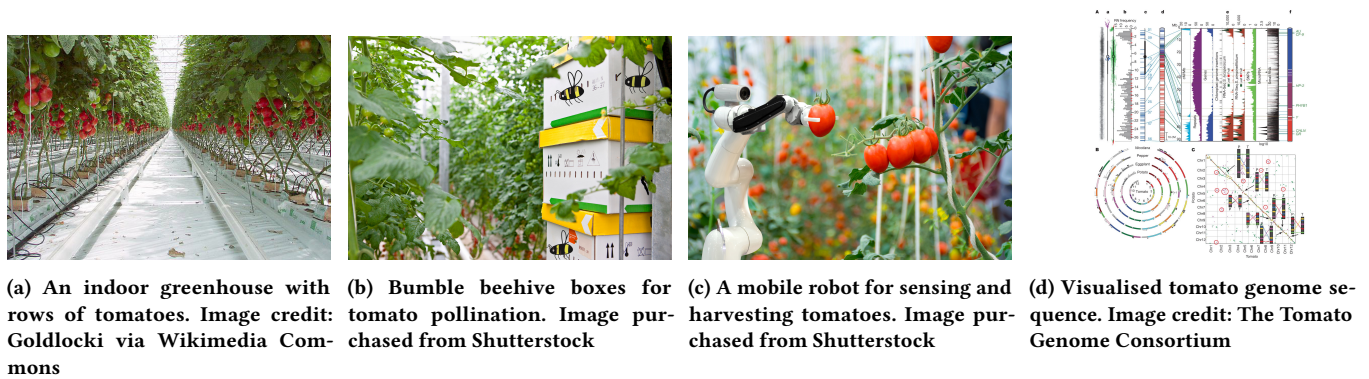


Figure 2: Indoor farm tomatoes at a university of agricultural research, the UK

a brief period, the greenhouse hosts bumblebees², which play an essential role in pollinating the tomato plants. All other insects are kept out, considered as pests in this tomato haven. The greenhouse maintains a consistent environment where temperature, humidity, CO₂ concentrations, water, nutrients, and light levels are carefully regulated by both human expertise and computer systems. This consistency ensures that the tomato plants are not subjected to any stress [1, 16].

Among the rows of cultivated tomato plants, mobile robots and numerous sensors serve as sentinels, capturing the journey of the tomato plant and monitoring the health, growth, and ripeness of its fruits from every angle. This meticulous attention to detail extends to the harvesting stage, where the uniformly ripe and red tomatoes are carefully picked, whether by a human harvester or a mobile robot³. With data collected from every stage of its life, and used to generate the seeds where its life is encoded, the plant and its fruits give rise to what could be described as digital twins - virtual counterparts that can simulate past, present, and future scenarios of its existence as a representational example of the same variety of tomato plant under controlled conditions [1, 16].

1.2 Heirloom tomatoes in a community garden, Italy (Figure 3)

In the heart of a community garden, a tomato plant proudly displays an heirloom tomato. This unique and plump tomato carries a story and lineage that spans generations of tomatoes, which have lived with and been tamed by humans for centuries. It has been cultivated naturally by farmers and gardeners who believe in the living power of heirloom tomatoes, using open pollination and the tradition of ‘seed saving’⁴. Unlike controlled hybrid or genetically modified varieties, heirloom seeds embark on a journey influenced

by diverse geographical locations. As the seasons pass, seeds transform into heirloom tomato plants, responding to unpredictable forces of nature, and human and other-than-human actions.

In the open garden, enriched with organic matter and compost, the heirloom tomato plant interacts with various humans and other-than-humans. Viruses, microbiomes, insects, worms, birds, and farmers play a role in the intricate ecosystem. Bees, butterflies, and gentle breezes aid in the tomato plant’s fruit production. Humans act as guardians, protecting the plant from potential threats and ensuring a fruitful harvest. They provide water and nutrient-rich soil while also removing unwanted plants to create an ideal sanctuary for the tomato plant to thrive. This resilient tomato ripens into a medley of colours - purple, red, yellow, and greenish - symbolising its strength and fresh flavour. It often surprises us with its diversity compared to the uniformity of greenhouse tomatoes. Perhaps it will grace the table of a local farmer, or its seeds will be preserved as a tribute to its capability and promise for endurance and an ever-evolving legacy [1, 19]

1.3 Wild tomatoes on a rocky hill, Galapagos (Figure 4)

In the untouched area of the Galapagos Islands, a wild tomato thrives in a wet and rocky area, embodying the untamed spirit of nature. The story of this wild tomato begins with a seed, a resilient lineage of native tomato species carried by the wind and animal guts and footprints. By chance, it finds itself nestled in this unique corner of the Galapagos with year-round sunlight and rain. As the seed germinates, the tomato plant becomes intricately entangled in the complex web of interactions and encounters that define the wilds of nature and persist for its survival. In this less human-impacted area, opportunities for cross-pollination with different tomato varieties are limited. Instead, the tomato plant self-pollinates, relying on the movements of endemic species and the wind. Adorned with spiky hairs, the tomato plant, and its fruit, stand as fortresses against the challenges of nature. These small, berry-like tomato fruits embrace marvels of adaptation passed down through generations. These evolving characteristics not only increase the plant’s chances of survival but also provide unusual resistance to diseases [29].

With the wisdom and knowledge of ancestors encoded in its genes, this wild tomato thrives without human intervention or

²Bumblebees are widely used for commercial pollination. This can lead to “a 30% increase in the yield of tomatoes, compared with pollination by hand, with a wand”, according to an article in the Guardian. <https://www.theguardian.com/environment/2019/may/04/the-battle-to-save-the-worlds-biggest-bumblebee-from-european-invaders>

³The harvesting robot-making company addresses that “greenhouse horticulture faces serious labour challenges, with scarcity of resource for crop maintenance and harvesting. Added to this, increasing expenses and reduced migration are leading to significant harvesting cost rises” <https://grow.ridder.com/>

⁴This traditional method has been maintained in agriculture for the last 1,200 years. Currently, grassroots local farmers also use this method to save open-pollinated and non-hybrid seeds. <https://seedsavers.org/preservation/#the-exchange>



(a) a community garden in Amsterdam. Image credit: Youngsil Lee



(b) a caterpillar and a tomato plant. Image purchased from Shutterstock



(c) various heirloom tomatoes. Image purchased from Shutterstock

Figure 3: Heirloom tomatoes in a community garden, Italy

protection. It grows its body and seeds, developing its own situated knowledge through resilient expressions of life. The tomato seeds holding the new knowledge need to be eaten by endemic creatures like giant tortoises, which can digest tomatoes and ferment the hard coated seeds. This helps break the extreme seed dormancy and effectively disperse them. A small number of these seeds may successfully germinate in different locations, perpetuating the cycle of life with various stories [26].

The life stories of these three tomatoes demonstrate their varying relationships with humans, other-than-humans, and their environment. These different ecologies influence the tomatoes in diverse ways and are, in turn, influenced by the tomatoes' embodiments and responses. In the next section, we delve into the diverse forms and characteristics of data materialised by both humans and other-than-humans, which affect how these tomatoes sense, embody, and evolve in different contexts.

2 FROM 'TAKEN' TO 'GIVEN' DATA AND 'PHENO-DATA': MANIFESTING THE ENTANGLEMENT OF MORE-THAN-HUMAN ECOLOGIES

Often likened to "new oil," data has been interpreted as a resource which is extracted for endless economic growth, particularly through the so-called rise of "big data" and "datafication" [4, 8]. Such interpretation has been widely critiqued by social scientists primarily due to its connection to the concentration of benefits by a few within Western industrialised contexts, which leads to social, environmental, ecological, and ethical concerns [4, 5]. Such critiques have also given rise to critical concepts such as data colonialism [7, 22], data capitalism [28], data surveillance [31, 33], and data extractivism [4]. Jansen, as cited by Becker [3], interestingly articulates this extractivist notion of data by discussing the word's etymology in the context of science and rationalism:

It is an unfortunate accident of history that the term datum (Latin, past participle of dare, "to give") rather than captum (Latin, past participle of capere, "to take") should have come to symbolize the unit-phenomenon in science. For science deals, not with "that which has been given" by nature to the scientist, but with "that

which has been taken" or selected from nature by the scientist in accordance with his purpose. [3, p.279]

In other words, rather than something that is 'given by' nature and should therefore be observed and attuned to, current notions of data are increasingly looking at it as something to be extracted, 'taken from' nature for maximal profit. In response to this, we focus on different expressions of livingness (i.e., the state or quality of being alive) of tomatoes in different environments, as a way to help us pay attention to data 'given' by nature or phenomena. Such expressions can be embodied into tangible forms and evolving characteristics of tomatoes by responding and "becoming with" [13] the situated environment. In our planet's ecosystems, living organisms, each with their own bodies, agency, and rhythms, are mutually influenced by intra-actions with various species and non-living matter like viruses, land, water, and machines [18]. Such intra-actions emphasize that distinct bodies and agencies don't precede relations but emerge through them instead. Specifically, the unique qualities and abilities of living organisms allow them to actively embody the entanglement of these more-than-human ecologies. On Earth, where everything is entangled, they create a temporal separation to bodily experience specific phenomena as "part of the world's dynamic engagement in practices of knowing". In her book, *Meeting the Universe Halfway*, Karen Barad [2] uses the example of brittle stars—brainless and eyeless creatures—to illustrate how living organisms can use their bodies for knowing and becoming with the world.

The brittlestar's bodily structure is a material agent in what it sees and knows as part of the world's dynamic engagement in practices of knowing. Similarly, its bodily materiality is not a passive, blank surface awaiting the imprint of culture or history to give it meaning or open it to change; its very substance is morphologically active and generative and plays an agentive role in its differential production, its ongoing materialization. [2, p.375]

In this paper, tomatoes are the material agents and living organisms that actively and evolutionarily adapt to processes of differentiation from distinct worlds. They continuously manifest the entanglement of local phenomena through their physical presence in situ, showcasing a variety of observable traits known as phenotypes.

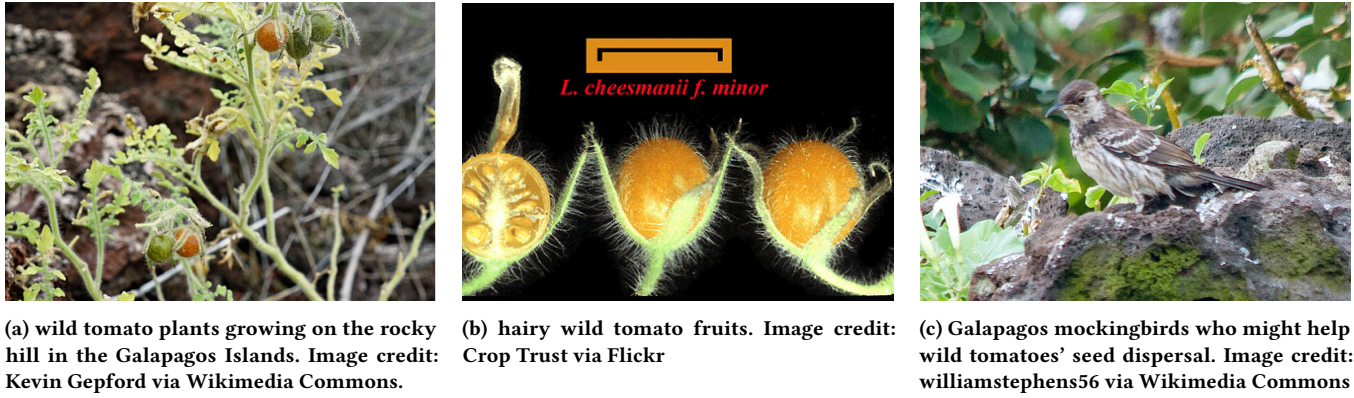


Figure 4: Wild tomatoes on a rocky hill, Galapagos

These traits are the result of intra-actions (a concept emphasising that distinct bodies and agencies do not precede relations, but instead, they emerge through them) between an organism's genes and its environment. This complex process enables organisms to develop different body structures, behaviours, and products (e.g., sound, light, and smell) within diverse environments, including the ability to change their DNA and pass on these traits through genetic inheritance [6]. Correspondingly, in our concept, data is seen as tangible, relational and 'becoming with' the changing environment, which we refer to as "Pheno-data" [18]:

Pheno-data combines the prefix "Pheno" derived from phenomena and phenotype, with "data", representing the entanglement of the world. Pheno-data therefore represents the livingness of the world tangibly, through evolving characteristics (colours, forms, movements, and rhythms) and responses (participation and engagement) of organisms which are equally shaped by and in turn shape the environment (other organisms, non-living matter, etc.).

The distinct Pheno-data from the three tomatoes clearly illustrate their unique 'embodiments' and 'responses' to differences and fluxes in more-than-human ecologies. These changes in temporary ecologies continue to incorporate data derived from social, cultural, and environmental phenomena of the more-than-human world, regardless of whether they are accessible to humans and their tools. Indeed, the view of all sentient beings can only be partial, as they notice and perceive a fraction of the entangled changes of the lifeworld through the Pheno-data.

In the following section, we will delve into the evidence of 'taken data', 'given data', and 'Pheno-data' by revisiting the three tomatoes' stories. The aim is to provide a tangible example of the introduced data concepts by situating them in "the intricate interdependence between humans, animals, plants, and other entities (viruses, machines, and the land) within ecological frameworks" [18].

3 VARIOUS DATA EMBODIED BY THE THREE TOMATOES

To summarise, in this paper, 'taken' data refers to data extracted from nature for a particular purpose [3], which is often aligned

with reductionist interpretations of data science [21]. These interpretations are in turn often embedded with anthropocentric and capitalist values and purposes, that is "ideas of progress and the spread of techniques of alienation that turn both humans and other beings into resources" [30, p.19]. In contrast, 'given' data is the collection of all existing and potential local manifestations in relation to an organism or phenomena in a specific environment. 'Pheno-data' refers to the manifestation and response of living organisms, as *becoming with* the environment. Let's revisit the three stories of the tomatoes and delve into these variations of data. This will help us better understand the meanings and impacts of these different data aspects on our tomatoes:

3.1 Expressions of data embodied by indoor farm tomatoes at a university of agricultural research, the UK

These tomatoes have complied with data dictated by computational systems and human experts, aiming to enhance the productivity and efficiency of humans, bees, soilless media, water, nutrients, machines, and the interior environment of the greenhouse. The 'taken' data in this context would be ⁵:

- Pre-loaded biodata in seeds for germination: Tomatoes' physical and genetic traits inherited from their parentage can be manipulated using breeding and genetic techniques to achieve various goals. These goals include delaying ripening, enhancing tolerance to biotic (viruses, bacteria, and fungi) and abiotic (drought, heat, cold, and salt in soil) stresses, improving fruit quality (including smell, taste, nutritional content for human health, appearance, and firmness), and promoting seed development.
- Computational data for growth in a controlled environment: Optimal amount of water and nutrients, temperature, light, air composition, CO₂, and artificial substrate; Bees (as temporary labour workers for pollination); humans (greenhouse

⁵These items were inspired by the book *Tomatoes*, written by Bioscientist Ep Heuvelink. This book offers a comprehensive and accessible overview of all aspects of tomato crop production.[16]

horticulture, economics, agrotechnology, statistics and breeding experts and harvest workers); sensors and mobile machines.

- Financial data: The yield of tomato production, the revenue generated from selling them, and the costs including electricity and water consumption for heating, humidifying, watering, data collection, growing system automation and optimisation, as well as the expenses for maintaining the facility and paying wages.
- The physical expressions of the tomato plant, its flowers that tell bees when to pollinate, or its ideal size or uniform red colour of its fruits that tell a human or robot when to pick it up.
- The specific classification and pricing of tomatoes for global markets.

In this controlled indoor environment, the tomato plant has no interactions with viruses, microbiomes, soils, and most insects. They have only occasional contact with humans (scientists, engineers, and facility cleaners) and other non-human entities (like bees, mobile robots), and no wind or weather changes. Its 'given' data could be:

- Reactions to designated location and direction for growth.
- Reactions to specific supplies of heat, water, CO₂, moisture, light, and chemical nutrients.
- Reactions to number of Bumble Bees introduced in the greenhouse
- Colour, shape, and number of germinated fruit
- By-products created by tomato plants, such as CO₂ (from the same species)

In this human-oriented circumstance, the 'Pheno-data' of commercial tomatoes is impoverished, it could be:

- Uniformity in appearance, growth and ripening cycle, and flavour.
- Predictable amount of tomato fruits.
- Ease of manipulation as a common model crop ⁶ for bioscientists.
- Flavourless and watery nature that make commercial tomatoes earn nicknames such as "waterbomb" [15] and considered "fake" ⁷, compared to locally grown tomatoes in European countries.

These tomatoes embody and respond to data primarily created by humans with profit-driven knowledge and purposes. In this economic-dominated system of greenhouses, the main challenge lies in the high energy cost required to operate them on a large scale for the global market while maximising profit. To address this issue, scientists are focusing on improving energy efficiency through further genetic manipulation. This includes investigating the possibility of tomatoes growing in the dark without the need for photosynthesis [11] or utilising tomato flower stem cells to grow fruits without the rest of the plant (stems and leaves) [27].

⁶Tomatoes are commonly used as a 'model crop' for diverse physiological, cellular, biochemical, molecular and genetic studies because they are easily grown, have a short life cycle and are easy to manipulate [16].

⁷This is based on the statement from Craig Lehoullier, a renowned author and gardener, about commercial and heirloom tomatoes: "You will realize that those round, pinkish-red things that are available in the stores bear no resemblance at all to the real tomato".

3.2 Expressions of data embodied by heirloom tomatoes in a community garden, Italy

These tomatoes are living in the open environment, adapting autonomously to environmental changes. Even though they are germinated, cultivated, and managed by humans for social, economic, and environmental purposes, they build their own resilience in this open environment. The taken data these heirloom tomatoes embody would include:

- Data for germination by humans: Seeds are commonly germinated in a greenhouse, which provides a stable and relatively warm, humid environment, to enhance the germination rate and boost their initial strength for high-yield production. This practice aids in their adaptation to environmental changes of an open field such as soil quality, moisture levels, air composition, temperature, and weather conditions.
- Data regarding the frequency and infrequency of various gardening tasks performed by humans, including but not limited to watering, composting, pruning, and weeding.
- Local farmers and gardeners consider these heirloom tomatoes, which have rich and diverse flavours, as "real" tomatoes.
- Data from human social and cultural phenomena: Geographical and social provenance to evidence food security with local resilience; accounting of associated activities and communities involving physical labour, social and cultural well-being; heritage seed libraries and seed banks and vaults ⁸; evidence of the retention of biodiversity within a region; documentation of associated use and recycling of waste/energy including carbon footprint; documentation of the stake that minority groups may have in the production cycle ⁹.

And the 'given' data could be:

- Reactions to whims of the weather.
- Chemical signals that symbionts such as viruses, microbiomes and fungi can grasp as an indication of optimal or suboptimal conditions for growth.
- Amount and shape of flowers that can attract pollinators (bees, butterflies, and flies).
- Withered leaves that indicate lack of water.
- Space taken by the plant itself, which indicate to other plants the need to grow in a different direction.
- Holes in leaves that indicate the presence of snails, birds, or other organisms feeding from the plant.
- Responses to other tomatoes or plants through natural open pollination.

In this open field where humans and other-than-humans coexist, the 'Pheno-data' of heirloom tomatoes we can perceive would be:

⁸Seed banks and vaults preserve plant genetic diversity and for future generations. The article stresses that gene banks are not a safeguard against apocalypse, as "organisms are what they are because of their genes and the effects of the environments." This implies that we cannot revive organisms without restoring the previous environment in which they thrived. <https://www.bostonreview.net/articles/jack-heinemann-seed-vault/>

⁹The item was inspired by the article titled *Can community gardens promote more sustainable lifestyles*. It discusses the potential benefits of community gardens for mental and physical wellbeing, as well as lifestyle sustainability. <https://www.chaireunesco-adm.com/13-Can-community-gardens-promote-more-sustainable-lifestyles>

- Variations in appearance, growth, ripening, and flavour, which evidence the different varieties and species of tomatoes and result in greater diversity, and complexity¹⁰.
- Temporal maturation in response to the plant's natural life-cycle and growth in different scenarios.
- Resistance and resilience in open fields depending on unpredictable conditions in a situated context.

Here, the relationship between heirloom tomatoes and humans can be seen as a form of companionship. These tomatoes have functioned as a living heritage throughout human evolution, fostering a bond based on "significant otherness" that extends beyond mere utility or domestication [12]. Heirloom tomatoes can thrive in open fields due to their robust resilience and receive protection from humans against threats like drought and harmful insects. Consequently, most of their seeds are preserved and germinated to perpetuate their diverse lineages and stories. In turn, humans derive mental and physical benefits by nurturing these tomatoes and enjoying their fresh and flavourful produce.

3.3 Expressions of data embodied by wild tomatoes in an untouched location, Galapagos islands

These tomatoes depend solely on their autonomous and agential livingness to thrive in the untamed wilderness. All the data they embody come from intra-actions with other-than-humans and environmental cues, constantly changing and evolving as part of the lifeworld. By being situated in the natural environment, wild tomatoes flourish through the unknown encounters and relationships that are closely tied to matters of life and death involving multiple species [30]. By paying attention to Pheno-data of wild tomatoes, we are able to perceive aspects that support the life of a wild tomato in the given environment. Here, data is only 'taken' by other-than-humans in their relationships with the tomato plant. The 'given' data, which wild tomatoes consistently embody while responding to other species and the environment, could be ¹¹:

- Data for germination: A rare moment for seeds to be eaten by seed dormancy breakers, such as tortoises, and be dispersed to suitable environmental conditions and locations to start a new life.
- Data for persistence, growth and evolution in the wild environment: All responses by sensing and embodying all intra-actions and encounters with living organisms and non-living elements in a specific location. By chance, a few small wild tomatoes can ripen from a tomato plant and encode their survival stories into their seeds. These seeds can then pass down their resilient and evolving legacy through generations, becoming part of the lifeworld.

In this wild where multi-species coexist, the 'Pheno-data' of wild tomatoes we can perceive could be:

- Naturally evolved distinctive characteristics, such as the berry-like, small, hairy orange-color fruits.
- Hard seed coats of wild tomatoes prevent germination in unfavourable environmental conditions and protect the seeds from invasion by microorganisms and mechanical stress [23].
- Survival in unpredictable conditions responding to "natural selection", enabled by biodiversity [24].
- Contingency of germination, survival, and growth of tomatoes.
- Resistance to plant diseases and fungal infestations common to tomatoes [32]
- Role within food cycles for endemic creatures such as Galapagos Mockingbird, Darwin's finches, and giant tortoises.

Wild tomatoes complete their entire lifecycle, from germination to the reproduction of future generations, autonomously. With the information inherited from their ancestors, they constantly intra-act with other-than-humans and the environment to embody situated knowledge, attuned to the contingency of nature. In this environment untamed by humans, living organisms dynamically reconfigure, regenerate, and recycle through their evolving bodies to sustain "multispecies' collaborative survival" [30] on this island.

In this section, we explored various types of data that influence the germination, growth, and evolution of these three tomato types in different environments. We also detailed how this data embodiment leads to their unique living qualities, referred to as Pheno-data. In essence, they exhibit varying degrees of autonomy and agency for life.

4 WHY DO WE NEED TO PAY ATTENTION TO THE AUTONOMY AND AGENCY OF TOMATOES' LIVINGNESS?

In the previous section, we explored different aspects of data in the context of three kinds of tomatoes within their different environments. The three approaches - taken data, given data, and Pheno-data, are created as lenses to illustrate how the tendency to focus on taken data can impoverish the other aspects of given and Pheno-data. In the story of commercial tomatoes, taken data plays a dominant role. It actualises human striving for control and creating sameness, consistency, stability, safety, improvement, isolation, and predictability as well as reducing complexity in tomatoes and its ecology. These intentions are obviously reflected in the control of the environment of commercial tomatoes and their consequent reduced capacity to give data or express its potential Pheno-data. In contrast, the stories of heirloom and wild tomatoes and their environments generate more difference, change, diversity, resilience, evolution, intra-action, and contingency with the given data of natural ecologies and Pheno-data of tomatoes with or without human intervention [17].

Though indoor farm tomatoes, heirloom tomatoes, and wild tomatoes all undergo a biological life cycle (germination, growth, and reproduction), and are therefore "alive" and "real", greenhouse tomatoes are often labelled as "fake" and "artificial", particularly in comparison to others, like heirloom tomatoes ¹². But what does

¹⁰The item was inspired by the book titled *Epic Tomatoes: How to Select and Grow the Best Varieties of All Time*. In his book, local gardener Craig LeHoullier describes his experience of growing heirloom tomatoes [19].

¹¹The items were inspired by the research of Charles M. Rick, an ecologist who studied the relationships between wild tomatoes and endemic creatures on the Galapagos Islands [29].

¹²Many people don't know about wild tomatoes, but wild tomatoes must be considered as real in the same context.

Texts from participants	Pheno-data	Pheno-fication
My lemon plant grew a bud today, perhaps it thinks it's spring due to my heated room	A growing bud from the lemon plant	- Sensitising: movement, temperature - Contextualising: Bud, lemon plant, heated room - Imagining: the plant might think it's spring
The gray to white feather transition on this goose indicate how it's growing into an adult, but could indicate environmental factors as well	Colour change of the goose's feather	- Sensitising: color transition of the feather - Contextualising: colors of feathers, aging goose - Imagining: a possibility of environmental influence on the feather
The bubble on the water is the result of bacterial breakdown of organic matter, and any disturbance of the substrate, be that by fish, wildfowl, or water movement	Bubbles from the water surface	- Sensitising: upcoming bubbles on the water - Contextualising: bacterial breakdown of organic matter, disturbance of the substrate - Imagining: thinking specific possibilities such as fish, wildfowl, or water movement

Figure 5: The relation of Pheno-data and Pheno-fication from the results of participants' assignment

"real" signify in the context of tomatoes? We argue that the distinction between real and fake originates from their autonomy and agency. These unique qualities allow living organisms to continuously respond to varying environments and evolve through diverse paths. In the case of tomatoes, these paths create a rich breadth of flavours that people often recognise as freshness or flavourfulness [19] and indeed "realness".

These stories about three tomatoes illustrate how the livingness of organisms can vary based on their autonomy and agency within different ecologies. In the ever-changing world, unique abilities are enacted to persist, repair, reproduce, and thrive by embodying and responding to the fluxes and differences. This is honed over multiple generations through trial and error, enabling them to remain as part of "the world's becoming" [2]. According to Barad, "Agency is a matter of intra-acting; it is an enactment, not something that someone or something has. It cannot be designated as an attribute of subjects or objects (as they do not pre-exist as such) [...] Agency is 'doing' or 'being' in its intra-activity." Further, she explains, "Agency is about changing possibilities of change entailed in reconfiguring material-discursive apparatuses [in our case, tomatoes] of bodily production, including the boundary articulations and exclusions that are marked by those practices in the enactment of a casual structure" [2, p.178]. We view the "bodily production" of living organism and tomatoes as autonomy in their autopoietic systems, which Donna Haraway describes as "self-producing" autonomous units that are only made possible within "sympoietic worldings" [14]. According to Haraway sympoiesis means "making-with". It refers to "complex, dynamic, responsive, situated, historical systems. [...] Sympoiesis enfoldes autopoiesis and generatively unfurls and extends it" [14, p.58].

Historically, humans as a species have developed a sense of appreciation (totemism) and responsiveness (a sense of safety and urgency) to the life and livingness of other-than-humans, who have brilliant resilience and adaptability [10]. This ability has been crucial for our survival in the unpredictable and untamed natural world. In today's world, however, we are undermining the autonomy, agency, and livingness of tomatoes, plants, animals, marginalised

groups, and nature itself. The reproduction of capitalist values leads us to view them merely as food and energy resources (using scientific measurements like g/ml/kWh), commodities (with price, weight, production origin, nutrition labels), labour (worker, pollinator, and cost), or hindrance (e.g. food waste, pesticide, and disease), driven by capitalist motives, limiting what we call data, which ignores the emergence of new forms of data and knowledge from them. Focusing on 'taken' data limits both humans' and other-than-humans' autonomy and agency to co-exist, respond, and connect with more-than-human ecologies, leading to lifeless worlds.

As biologist Lynn Margulis noted, "Life on Earth is more like a verb. It repairs, maintains, recreates, and outdoes itself" [20]. The level of control which is present in the commercial tomato story may lead the plant to generate more 'nouns' from a human-centric perspective: things, objects, numbers, texts, colour codes, costs, and revenues. Conversely, the data in stories about heirloom and wild tomatoes can lead the tomato plant to generate a greater range of 'verbs', signifying actions, responses, and changes, which strongly connect with life concepts like birth, interaction, persistence, coexistence, embodiment, evolution, and death.

How can we pay attention to the autonomy, agency, and livingness of tomatoes and restore our qualities for appreciation, sensuality, and emotionality [25] that can help us connect with them and the lifeworld? In the following section, we showcase a tangible way to experience the concepts of "Pheno-data" and "Pheno-fication" through design practice. Through fabulation workshops, we explored their potential for shifting perspectives from an anthropocentric to an ecological viewpoint.

5 FABULATION: EXPLORING PHENO-DATA AND PHENO-FICATION THROUGH DESIGN PRACTICE

We organised fabulation workshops with various groups of people, including farmers, scientists, and designers, to help us re-frame data towards Pheno-data. This approach, called "Pheno-fication," involves a process of sensitising, contextualising, and imagining

in order to notice and resonate with the knowledge and practices of living organisms in different environments. For example, as an assignment, we asked participants in a workshop to explore their surroundings and find data they believed was ‘given by’ nature. Figure 5 illustrates a few examples that participants brought as results, as well as their interpretation of pheno-data and the process of pheno-fication:

Participants then created stories which illustrate how a living organism might live in a recontextualised ecological world by centering a living organism’s evolving body and responses towards the world and mapping out their possible relationships with humans, other species, and social, technological, and environmental phenomena in a specific location (participants chose human-oriented contexts such as digital society, tourist area, city). This approach revealed “tensions between more-than-human knowledge and anthropocentric knowledge in imagined alternative lifeworlds” [18] by reconfiguring more-than-human pasts, presents, and futures shown at Figure 6. For instance, participants created stories where snails became urban heat indicators (substituting scientific measurement) and butterflies became a religion to worship (replacing gods) while Mockingbirds, with their ability to mimic sounds turned into Tind-birds (replacing sound technology). These instances demonstrate the livingness of organisms as more-than-human data and knowledge that have the potential to bring unknown possible worlds, even within the confines of human societal boundaries, systems, and knowledge [18].

Through the series of design workshops, we observed that participants tended to explore unseen influences, causalities, relationships, and flows that exist in the natural world. They also acknowledged the role of ecologies in defining the plurality and locality of Pheno-data of living organisms across various times, spaces, and scales. Specifically, the exploration of Pheno-data led participants to the emergence of a new realm of questions and imaginings (such as “Why did my lemon plant bud in my heated room? Could it be because the plant thinks it’s spring?” or “Why are there bubbles on the water surface? Could it be due to bacterial breakdown of organic matter, or the breath of fish or wildfowl?”) which would not be asked otherwise. It also helped to promote curiosity, appreciation, and responsibility towards other-than-human worlds [18].

In the specific context of tomatoes, Pheno-data and Pheno-fication allow us to become more aware of the changing physical characteristics of tomatoes and contextualise the matters and meanings of their relationships and participations, while imagining potential evolutions of tomatoes and multispecies world-makings [18]. With fabulation, the tangible designerly example would be like the life stories of the three tomatoes introduced at the beginning.

6 LET’S RETHINK HCI DESIGN PRACTICES WITH PHENO-DATA GIVEN BY TOMATOES AND LIFE WORLDS

In a review of epistemological approaches of SHCI, Disalvo et al. [9] state that “A move away from an emphasis on technology design, or even from research aimed primarily at informing such design, presents challenges to HCI as a field” and raise a critical question “if technology is not the point, then what becomes the work of sustainable H“C”T?” In response, we argue that the work of (S)HCI needs to

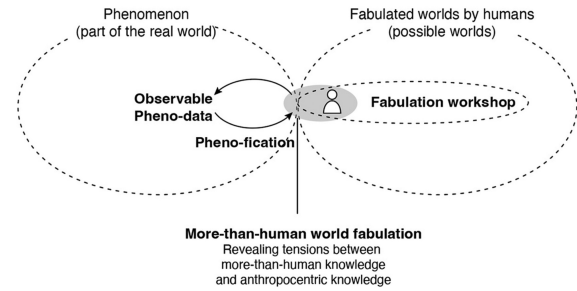


Figure 6: Visualised relations between Pheno-data, Pheno-fication and fabulation for a more-than-human world-making process

focus on fundamental inquiries into the ontology of data, which has been shaped (and limited) by anthropocentric and capitalist purposes and knowledge and been embedded in the design solutions of technologies in the HCI field.

In this paper, we use the case of tomatoes to question the ability of current interpretations of data as in ‘digital technologies’ to include other-than-human organisms beyond the purpose of generating global-scale, profit-oriented, and human-centered knowledge [18]. By asking only “which tomatoes are good and valuable for us?” without considering “what is a good life for tomatoes as living organisms,” or “how does the livingness of tomatoes that we support affect the livingness of other-than-humans around them?” we reinforce dominant capitalist perceptions of living organisms that side-lines their autonomy, agency, and ultimately whole ecosystems around living entities. The impact of this tendency to ask utilitarian rather than ecological questions extends beyond tomatoes to all other-than-humans, including animals, plants, water, and land, and may result in greater commodification, exploitation, and ecological impacts such as climate change, as well as inequality and discrimination towards other beings.

To emphasise the significance of these inquiries, in this paper, we have looked at three different varieties of tomatoes that have evolved differently by embodying data from human and other-than-human knowledge practices. Greenhouse tomatoes have been manipulated, controlled, and become “Frankensteined” tomatoes [1]. Heirloom tomatoes have adapted to human society and become “Companion” tomatoes. Wild tomatoes have actively participated with their own autonomy and agency in ecological matters and exist as “Living” species, sustaining the lifeworld through sympoiesis. As designers, we have also introduced the concept of “Pheno-data” as a thought-provoking idea and proposed “Pheno-fication” as a means to sensitise, contextualise, and imagine such data, which ultimately represents the knowledge of living organisms, ‘becoming with’ the world.

We envision that Pheno-data from more-than-human ecologies (e.g., the budding of a lemon plant, the colour changing in a goose’s feather, varied bird sounds in different geographical locations, dragonflies flying lower right before rain, bubbles appearing on the water surface in a reserve, and bioluminescence from microorganisms [18]) can facilitate further discussions and practices to critically rethink our relationships with the world and with data

in the context of more-than-human ecologies. We hope that this paper will invite the HCI community to reconsider data and data practices, and encourage them to explore new ways to expand the ecological dimension of our knowledge and reconnect with the life of the world.

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